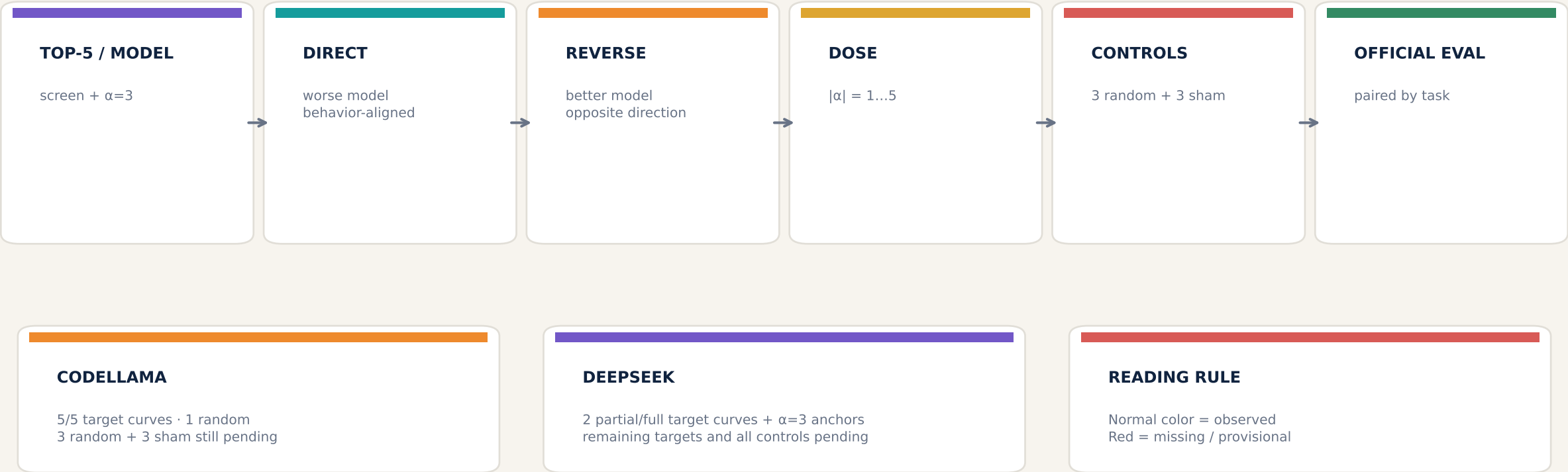


BBASV TURNS ASSOCIATIONS INTO CONTROLLED CAUSAL TESTS

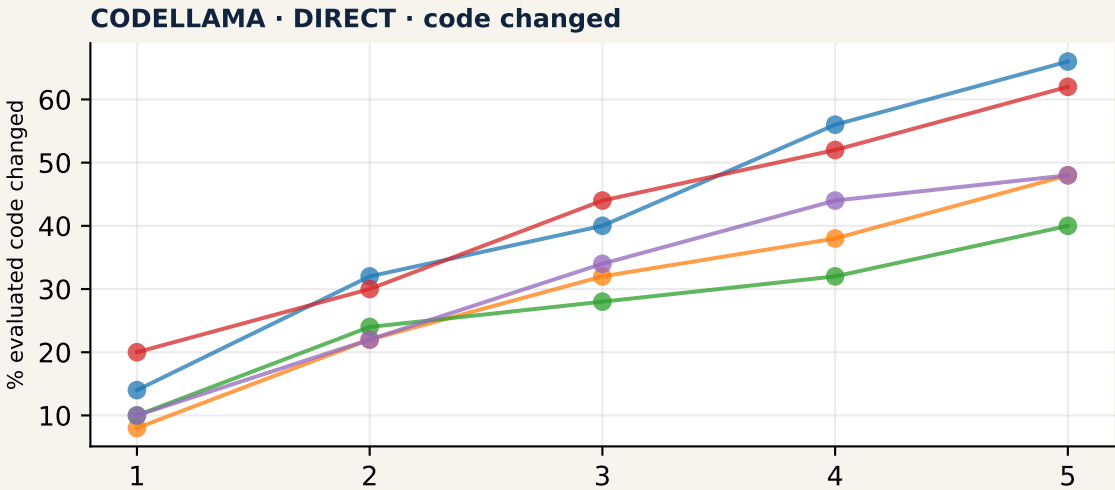
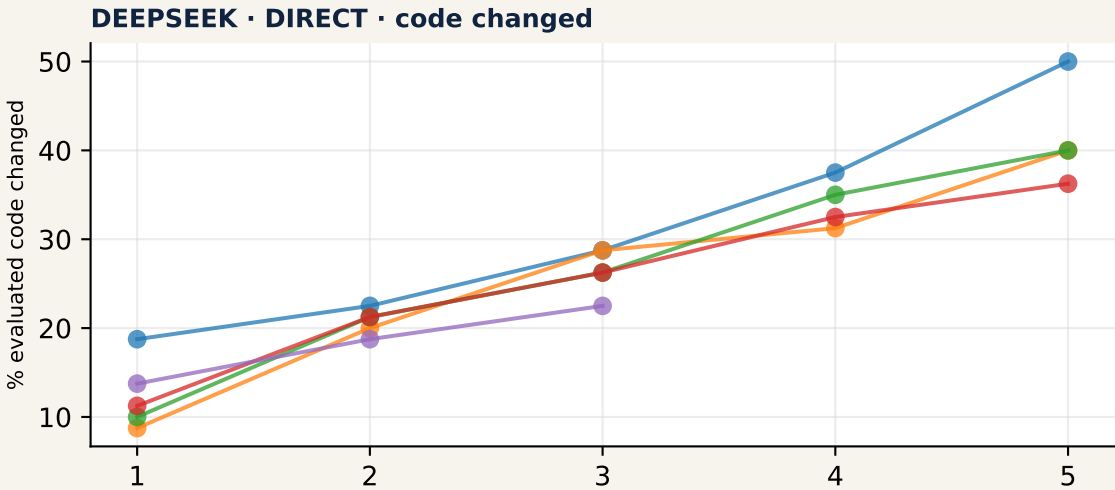
Bidirectional Behavior-Aligned Steering Validation



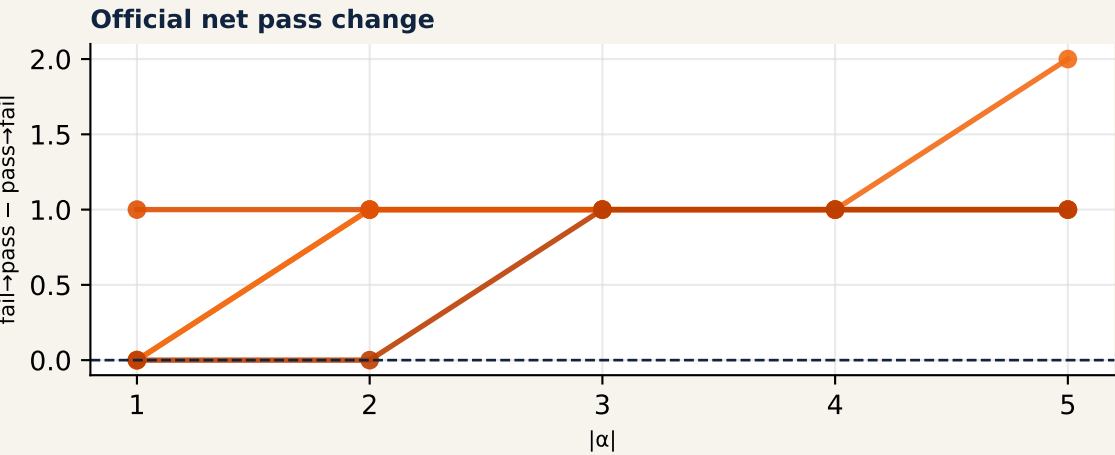
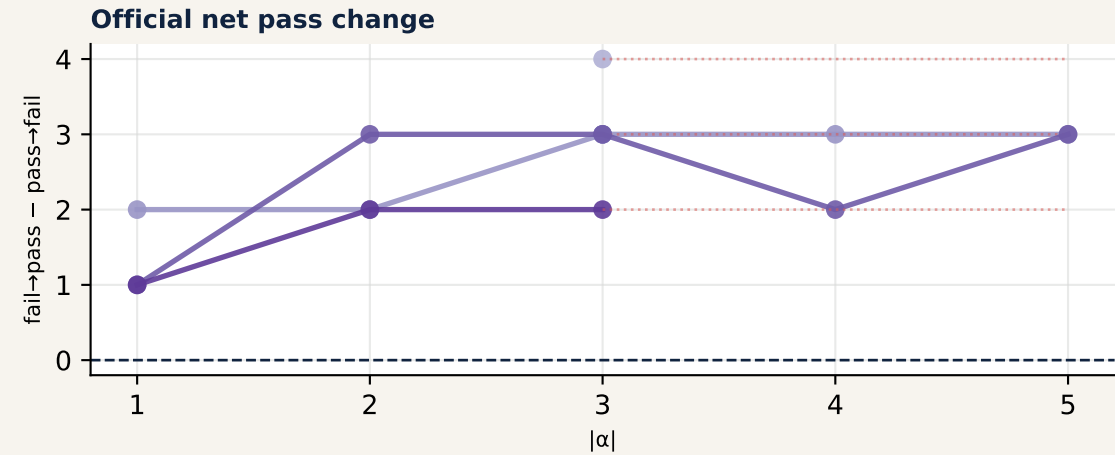
STEERING CHANGES TRAJECTORIES MORE OFTEN THAN OUTCOMES

Two aligned endpoints: evaluated-code change and official fail↔pass

TRAJECTORY EFFECT

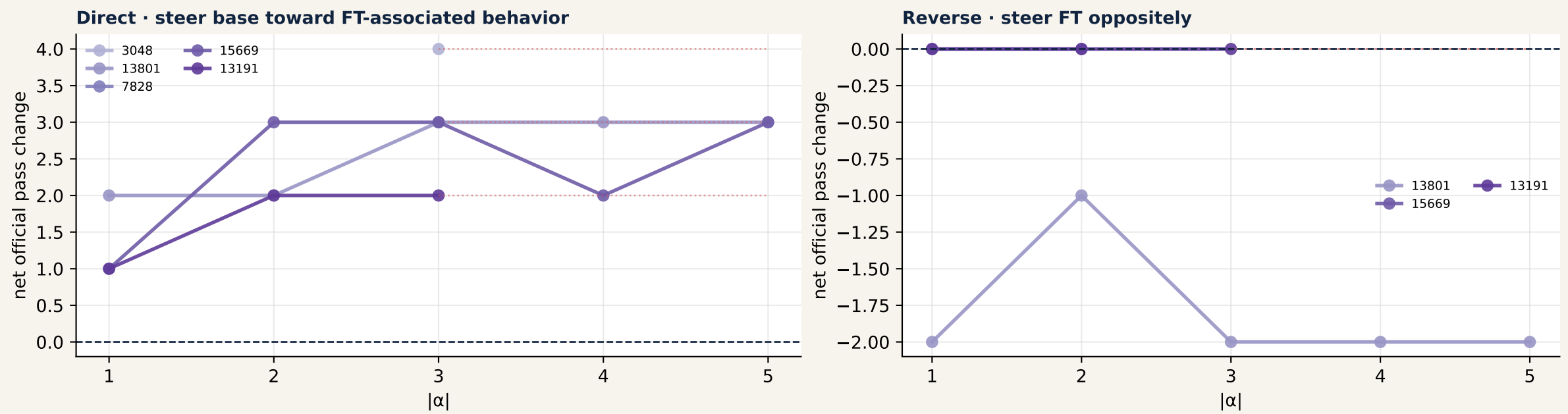


TASK OUTCOME



DEEPSEEK: EARLY RESULTS SHOW POSITIVE BUT TASK-CONCENTRATED

Direct baseline 15/80 · reverse baseline 57/80 · 25/45 non-baseline target arms evaluated



FAVORS THE STORY

13801: +2,+2,+3,+3,+3 direct
and -1/-2 net in reverse

QUALIFIES IT

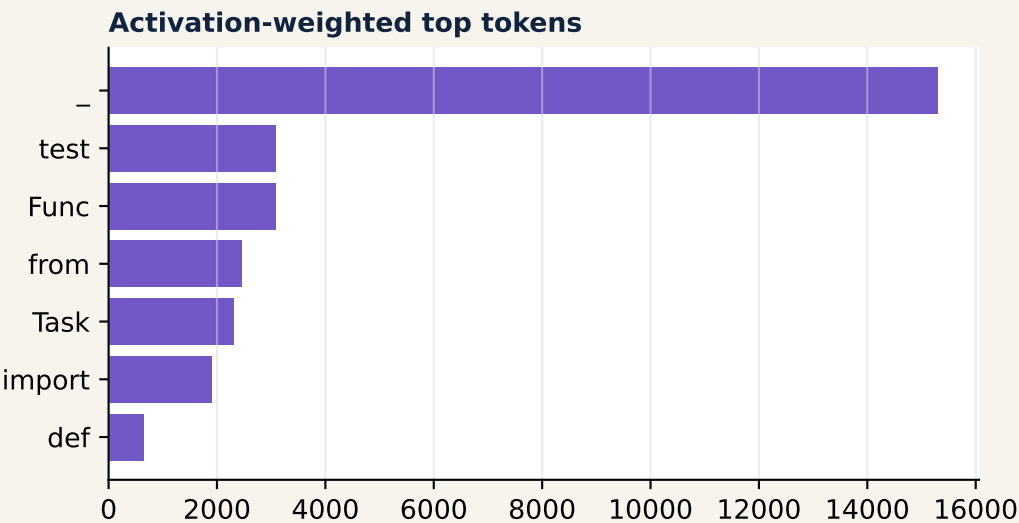
/1030 and /706 recur across targets:
part of the gain is task susceptibility

PENDING

3048 and 7828 dose curves
all random and sham controls

DEEPSEEK FEATURE 3048: A COHERENT TEST/IMPORT-BOUNDARY CAND

Semantic evidence and one clean $\alpha=-3$ fail→pass example



BIGCODEBENCH/762 · FAIL → PASS

BASE: broad try/except + appended pytest suite

STEERED $\alpha=-3$: direct implementation + return zip name;
post-solution tests removed

Official transition: fail → pass

MEANING

test/import/function-boundary contexts

FAILURE MODE

post-solution contamination and
over-defensive implementation

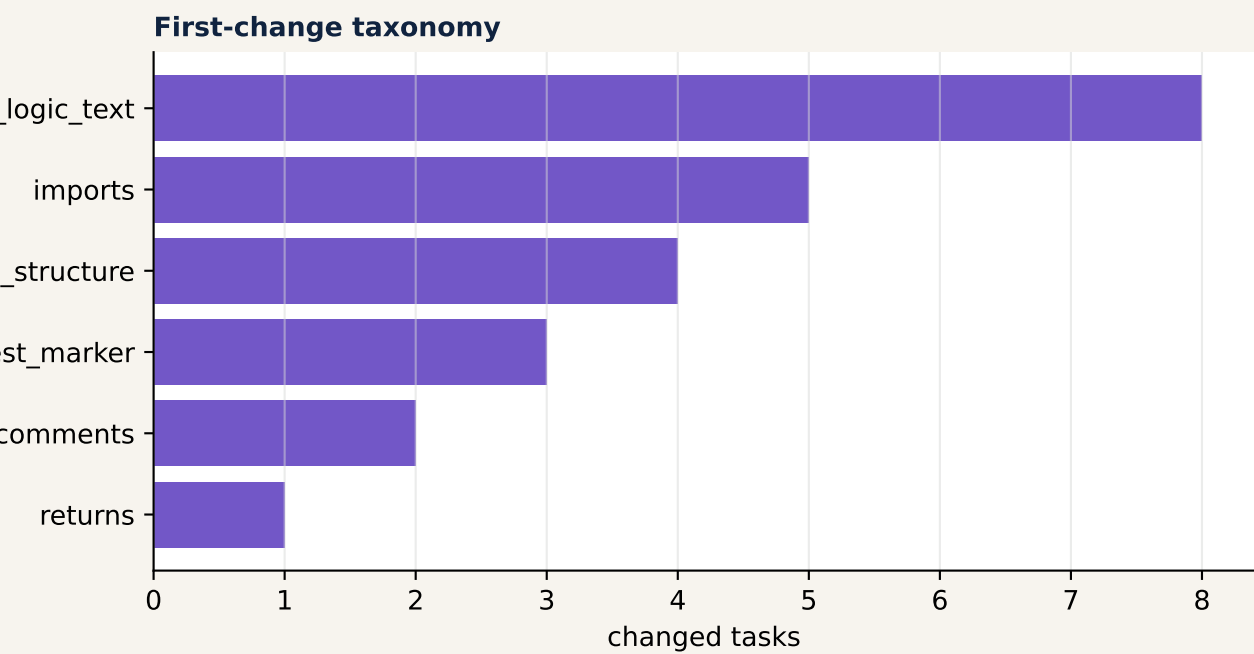
CAUSAL CHANGE

removes tests and simplifies body
in a task that becomes correct

Interpretation is coherent, but 3048 specificity versus controls remains PENDING.

FEATURE 3048 OFTEN CHANGES THE RIGHT REGION WITHOUT SOLVING

At $\alpha=-3$, 23/80 evaluated codes changed; only 4 became official passes



WHAT CHANGED

imports/tests/comments/function structure
frequently move—the expected semantic neighborhood

WHY MOST STILL FAIL

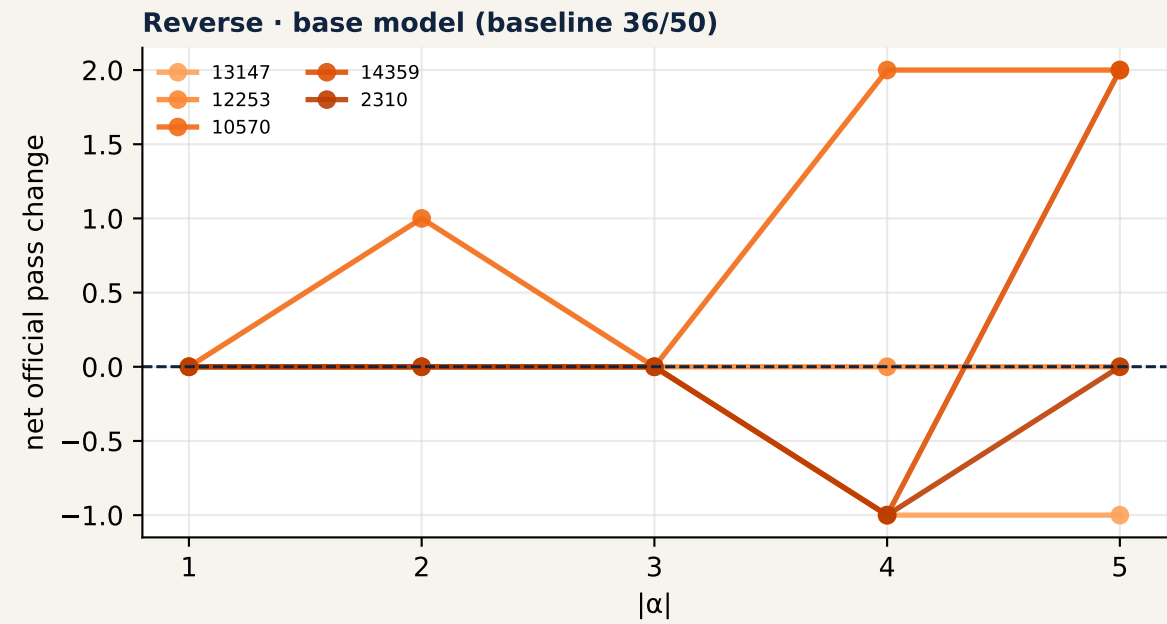
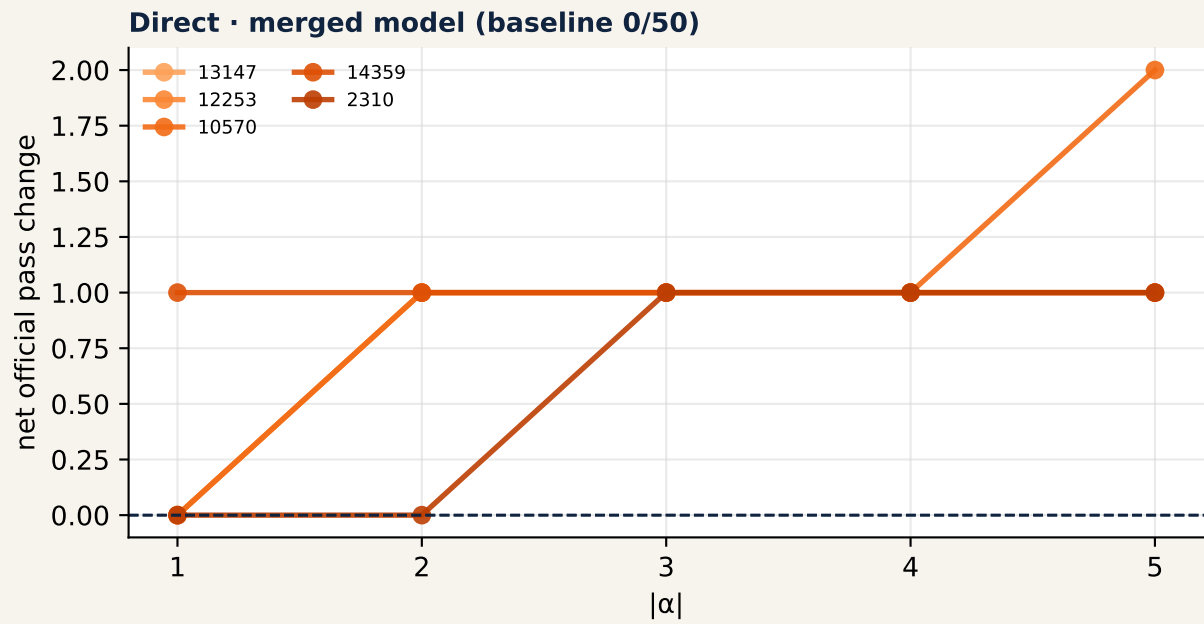
Removing contamination cannot repair every
underlying logic/API/runtime defect

23 code changes → 4 fail→pass

Causal scope ≠ complete task solution

CODELLAMA: COMPLETE TARGET CURVES SHOW WEAK OUTCOME SPEC

All 50 target direct/reverse arms evaluated · one random latent evaluated · shams pending



TARGET RESULT

mostly 0 or +1; 10570 reaches +2 at $|\alpha|=5$

CONTROL RESULT

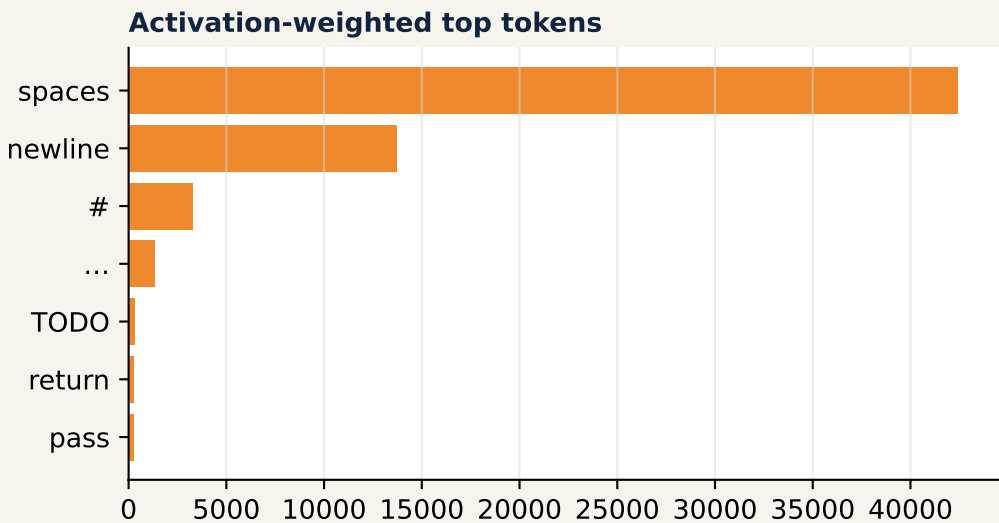
random 13879 reaches +1 direct and +2 reverse

CURRENT READING

trajectory control exists; outcome specificity is not established

CODELLAMA FEATURE 13147: A CLEAN RETURN-VALUE REPAIR

Task /490 is corrected from $|\alpha|=2$ through 5 and differs from the common /119 response



BIGCODEBENCH/490 · FAIL → PASS

BASE: parses XML and writes JSON—but returns nothing

STEERED: adds `return result` to both generated functions

Stable at $\alpha=-2,-3,-4,-5$

MEANING

incomplete function / placeholder /
return-boundary contexts

FAILURE MODE

required return value omitted

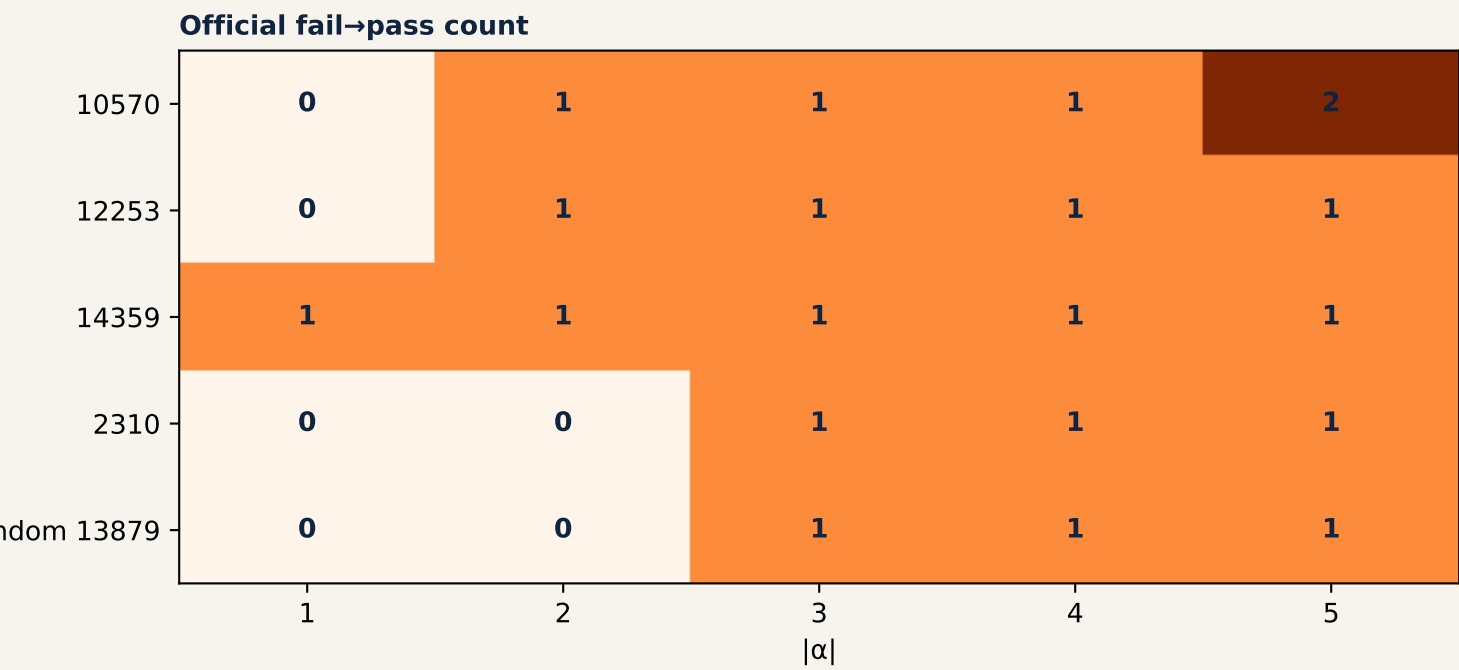
REVERSE TEST

base loses 1 pass only at $|\alpha|=4-5$

Promising mechanistic example; matched random/sham comparison for /490 remains PENDING.

CODELLAMA ALSO EXPOSES A NON-SPECIFICALLY STEERABLE TASK

/119 is corrected by four targets—and by the random feature 13879



AGAINST SPECIFICITY

A random latent reproduces the same /119 transition at $|\alpha|=3-5$

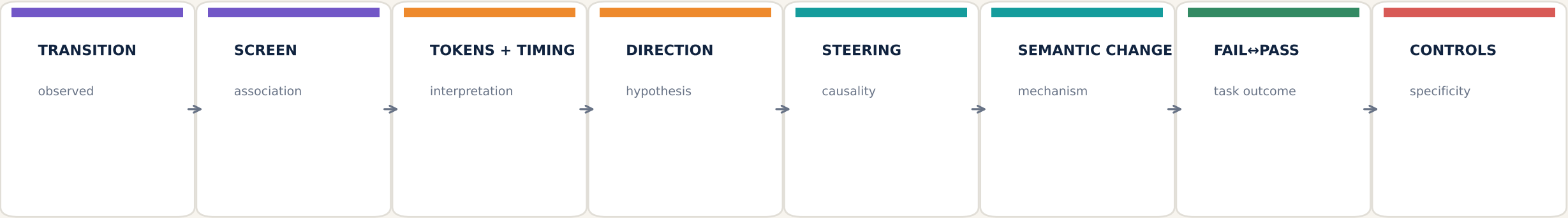
FOR THE METHOD

The control catches exactly the overclaim BBASV was designed to detect

Causal change? YES Feature-specific mechanism? NOT YET

THE EVIDENCE CHAIN MAKES EACH CLAIM AUDITABLE

Association, interpretation, causality, and specificity are different evidential layers



DeepSeek 3048 ✓ ✓ ✓ ✓ ✓ ✓ ✓

controls pending

CodeLlama 13147 ✓ ✓ ✓ ✓ ✓ ✓ ✓

partial controls

The methodology does not treat every fail→pass as feature-specific; it tests where the evidence chain breaks.

WHAT THE PARTIAL RESULTS SUPPORT—AND WHAT THEY DO NOT

Current claims are deliberately tiered

SUPPORTED NOW

- screening compresses 16,384 latents to testable candidates
- selected features causally alter generation trajectories
- some interventions create official fail→pass transitions
- 3048 and 13147 have semantically coherent examples
- controls identify task-level susceptibility

NOT YET SUPPORTED

- all model regressions/improvements are feature-caused
- every successful target is feature-specific
- CodeLlama targets outperform generic perturbations
- generalization beyond selected cohorts
- robustness across seeds

Best current causal claim

Selected CrossCoder features can exert directional, semantically coherent control in selected tasks.

Specificity is a result to be demonstrated—not assumed from screening or a single correction.

A REUSABLE METHODOLOGY FOR CAUSAL MODEL DIFFING

The output is a falsifiable mechanism hypothesis—not merely a feature ranking

1 Paired behavioral transitions

2 Same-text sparse representation

3 Conditioned multi-cell screening

4 Meaning + timing + failure mode

5 Directional causal hypothesis

6 Dose-response steering

7 Official paired evaluation

8 Random + matched sham controls

9 New seeds / held-out validation

Each stage can reject or refine the hypothesis before a broad mechanistic claim is made.

FROM MODEL DIFFERENCES TO TESTABLE CAUSAL MECHANISMS

The partial BBASV results already show both the promise and the discipline of the framework

Model differences are not only observable in behavior.

They can be localized in a shared latent representation,

interpreted in context, and—in selected cases—

causally manipulated.

THE CONTRIBUTION

A testable path from behavioral contrast → sparse feature → semantic hypothesis → controlled intervention → specificity test

PENDING: complete DeepSeek controls, CodeLlama sham/random controls, and multi-seed validation.